

Offer #SDS24-01F

Sežana, 2025-04-28

Axelered Doo

Partizanska Cesta 82, Sežana 6210, Slovenia

VAT no: SI71451331

Reg.no: 9247238000

Real Comm Srl

Via Dei Serviti 12, Porcia 33080, Italy

VAT no: IT01410160939

Description	Unit price	Quantity	Price
Smily Human Pose Software – Phase 2.0 Pre-prototype	€ 18,000.00	1	€ 18,000.00
Smily Human Pose Software – Phase 2.a Prototype	€ 14,000.00	1	€ 14,000.00
Smily Human Pose Software – Phase 2.b Production Software	€ 24,000.00	1	€ 24,000.00
Smily Human Pose Software – Phase 2.c Review 1	€ 2,000.00	1	€ 2,000.00
Subtotal			€ 58,000.00
VAT Reverse Charge VAT is not settled under Article 44 of Directive 2006/112/EC			€ 0.00
Total Price			€ 58,000.00

The quote is valid for 10 days from the date of issue, after the quote is rendered void and prices may vary for any reason.

State Of the Art

The Client is a manufacturer of hardware solutions designed for dental practitioners (referred to as “Workstations”). The Client has previously developed a prototype software capable of estimating the posture of a person standing in front of a camera. The prototype consists of a depth camera (OAK-D Lite) and a GUI (developed in Python, compatible with modern Windows desktop PCs). The camera is intended to be mounted on the workstation to detect the practitioner’s posture and provide real-time feedback on correct and incorrect positions. The Client has also sourced a new depth camera (Orbbec Femto Bolt), which will be integrated into the prototype software before the execution of the current service.

Nature of the service

The service involves the redesign and reimplementation of the current state-of-the-art (SOTA) system to incorporate additional features and make the product production-ready. The final software will be delivered as an installable application for Windows desktop PCs.

Delivery and Deliverables

The service provided by the Provider to the Client will be carried out in accordance with the documentation herein and will result in one or more deliverables, as specified in the table below. The Provider’s obligations are limited to the deliverables explicitly listed in this document; no additional deliverables are included unless expressly agreed upon.

Deliverable	Delivery Date	Notes
Pre-prototype	Delivered	Already delivered alpha version of the software.
Phase 2.a	28/11/2024 30/04/2025 30/05/2025	Beta version of the software.
Phase 2.b	28/03/2025 31/05/2025 04/07/2025 15/06/2025 (*)	Final version of the software.
Phase 2.c	28/07/2025 15/06/2025 (*)	After production delivery, the Provider and the Client will collaborate for this period to fix any bugs and unexpected behaviours in the software.

* Due to the urgency and short deadline, the Provider shall not be held liable for any delay in the delivery of the deliverables listed in this agreement.

Each phase introduces specific features and improvements to the software, with technical implementation details documented in **Annex A - Technical Documentation**.

Phase 2.a – Foundational Enhancements

This phase focuses on core improvements to the posture evaluation capabilities and foundational calibration mechanisms. The key deliverables include:

- Enhanced detection of multiple joint positions and angles.
- Support for exercises composed of multiple sequential postures.
- Calibration features for absolute measurements relative to the workstation.
- Automatic alignment and camera calibration, including calibration relative to the Workstation and mannequin positioning.
- System-generated warning upon detection of unacceptable camera positioning

Phase 2.b – User Experience and Reporting

Building upon Phase 2.a's foundation, this phase introduces interface enhancements and advanced reporting tools. Key deliverables include:

- Visual improvements in posture feedback (e.g., colour-coded error indicators, GUI overlays).
- Configurable start/resume countdowns for exercises.
- Option to disable real-time feedback for exam-mode scenarios.
- Exportable advanced reports, including error logs, timestamps, and snapshots.
- Posture editing interface for customizing evaluation constraints.

Phase 2.c – Maintenance and Stabilization

In the final phase, the Provider and Client will collaborate to stabilize the software and address any bugs or unexpected behaviour encountered after deployment.

Exclusion

This quote is intended for the delivery of what is listed in the section "Delivery and Deliverables" and is further described in other sections of this document. Any feature, service, or delivery not explicitly listed herein shall be deemed excluded. For clarity, the following non-exhaustive list highlights some notable exclusions:

- Modifications, updates, maintenance, testing, or debugging of any code that has not developed by the Provider.
- Any documentation other than the general documentation of the delivered deliverables.
- Any other written material, graphics, presentations, explanation, or RFC intended for any use.
- Training or TT (technology transfer) to any individual or organisation that has not been explicitly listed and agreed to within this Document.
- Maintenance, modification, or updates of any deliverable, beyond the point where correct working of the deliverable is assured with the current state-of-the-art, or in any case after the third month since the delivery of the individual deliverable.

Client Deliverables

To fully complete the project as outlined, certain deliverables are still required from the Client. The timely provision of these items is essential for the proper configuration, calibration, and finalization of the software components. Should the Client fail to deliver the required materials, data, or specifications within a reasonable timeframe, the Provider cannot be held responsible for any resulting limitations in functionality or incomplete features. In such cases, the Provider reserves the right to consider its contractual obligations as fulfilled based on the deliverables completed up to that point. The Client is expected to deliver:

- Complete reference exercise to be integrated in the Phase 2.a software including all parameters that fully describe the exercise as detailed in **Annex A**.
- Following the delivery of each deliverable, the Client shall conduct prompt and thorough testing, providing exhaustive and detailed bug reports including all necessary reproduction details.

Attachments

This quote references the following attachment, which forms an integral part of the agreement and defines the technical scope of the deliverables:

- **Annex A:** Technical documentation Rev03

Only the features, behaviours, and specifications explicitly described herein or in the annexes are to be implemented. By accepting this quote, the Client confirms having reviewed the annexes in full and acknowledges its accuracy, completeness, and suitability for the intended use. Any additional requirements or changes not expressly included in the annex will be considered out of scope and subject to a separate agreement or change request.

THE AGREEMENT

1. **Agreement.** This agreement, when duly signed by both parties, becomes a legally binding Sales Agreement and is entered into on the last date of signature (the “Effective Date”). This agreement, by and between Axelered Doo with a VAT number of SI71451331 and an address of Partizanska Cesta 82, Sežana 6210, Slovenija (the “Provider”) and Real Comm Srl with a VAT number of IT01410160939 and an address of Via Dei Serviti 12, Porcia 33080, Italy (the “Client”), also individually referred to as “Party”, and collectively “Parties”.
2. **Scope.** This Agreement governs the sale of a service by the Provider to the Client. The outcome of this performance will be what listed “delivery and deliverables” that will be duly delivered by the Provider to the Client in the form and within the terms described within the same section, and better described in other sections of this document. Any other activity is excluded with particular attention to what is listed in the “exclusions” section.

COLLABORATION & ACCESS

3. **Collaboration.** Both parties agree that given the nature of the service the Provider service strongly relies on Client collaboration, who should promptly reply to any Provider inquiry and offer any reasonable support needed to use and understand the current state-of-the-art.
4. **Systems.** The client should grant the Provider any required access to technologies and architectures (Systems) owned or controlled by the Client and/or any third party on its behalf. Such a System shall remain in function and accessible to the Provider for the entire duration of the performance.
5. **Access to state-of-the-art.** When existing, the Client should grant Provider access to any asset and knowledge related to the current state-of-the-art (including but not limited to code, documentation, or design) and should provide any additional information or infrastructure necessary to use, understand, or test the current state-of-the-art. This includes direct contact with anyone who has been involved with the conceptualization, design, or realisation of the state-of-the-art.
6. **Third-party systems.** Any third-party system required for the performance of the service should be obtained by the Client and made accessible to the Provider for the entire duration of the performance, free of charge. The Provider will not pay for, nor subscribe-on-behalf-of, for any such system. Any such system that may be required has been mentioned in this document (unless otherwise explicitly stated) and the Client herein declares to have evaluated and be aware of such expenses.

INTELLECTUAL PROPERTY

7. **Intellectual Property.** The result of the performed service will remain the Provider’s intellectual property until the client performs the due payment in full. Both parties agree that after such an event the Provider will release the intellectual property to the Client who will become the sole owner.
8. **Third-Party Intellectual Property.** If the Provider uses or licences intellectual property owned by third parties in the performance of the service, it is the Provider's responsibility to inform the Client who shall verify the licence terms and obtain and maintain any such licences and authorizations as necessary. This condition also applies to open-source intellectual property licensed under any licence.
9. **Confidentiality.** Each Party shall maintain confidentiality of all such confidential information, and without obtaining the written consent of the other Party, it shall not disclose any relevant confidential information to any third parties, except for the information that: (a) is or will be in the public domain (other than through the receiving Party’s unauthorised disclosure); (b) is under the obligation to be disclosed pursuant to the applicable laws or regulations, rules of any stock exchange, or orders of the court or other government authorities; or (c) is required to be disclosed, by any Party, to its shareholders, investors, legal counsels or financial advisors regarding the transaction contemplated hereunder, provided that such shareholders, investors, legal counsels or financial advisors shall be bound by the confidentiality obligations similar to those set forth in this Section. Disclosure of any confidential information by the staff members or agencies hired by any Party shall be deemed disclosure of such

confidential information by such Party, which Party shall be held liable for breach of this Agreement. This Section shall survive the termination of this Agreement for any reason.

PAYMENT

10. **Fixed Price.** When applicable, The Fixed Price quote implies that no additional cost beyond the quote can be charged by the Provider in case the quoted price for the Service does not reflect the actual costs faced during the performance of the Service. In the event of any Modification to this agreement or any of the requirements mentioned herein, the Fixed Price shall be waived or renegotiated. For the Price estimate, the Provider has relied on information provided by the Client (or other authorised personnel on its behalf); in case any such information is not accurate or not true any Provider obligations regarding the Fixed Price shall be deemed inapplicable and the Provider may charge the Client with any additional cost faced during the performance of the service.
11. **Price.** The Provider agrees to sell to the Client the service for the price listed in the “offer” section. The Client accepts the sale and understands that due to its nature, the performance of the service is not refundable and in any case shall be paid.

MODIFICATION & TERMINATION

12. **Modification.** No modification or alteration of this contract shall be considered as having been made unless done with consent by the Parties and fully executed in writing and duly signed by the Parties. Therefore, no modification can be made to the deliverables or the project specification unless explicitly agreed by both parties.
13. **Termination.** This Agreement may be terminated at any time by either Party upon written notice to the other Party. The Client will be responsible for the payment of the service that has already been performed by the Provider. In case of termination, the due amount will be the full price of any completed deliverable and will be computed “pro rata temporis” for any uncompleted deliverable. This also applies when the outcome of the service is partial, rendered unusable, or lacks the properties described in “delivery and deliverables” due to termination.

OTHER

14. **Service start.** The service will start on the latter of this contract being duly signed by both Parties, all advanced payment being received by the Provider, and both Parties have agreed to start the Service and fully commit to ensuring its successful delivery.
15. **Delivery terms.** The Provider will deliver the deliverables within the terms as described in the section “delivery and deliverables”. Different deliverables or alterations of the same (such as, but not limited to, lexical translations, change of programming language or optimizations) are not due unless explicitly agreed hereunder. Unless differently specified all the delivery terms are expressed in weeks counting since the Service start date.
16. **Excuse for Delay or Failure to Perform.** The Provider will not be liable to the Client for any delay, non-delivery, or default of this agreement due to labour disputes, delay, shortage of systems/services/goods required to perform the Service, malfunctions, fires, accidents, Acts of God, or any other causes outside of the Provider’s control. The Provider shall notify the Client immediately upon realization that it will not be able to perform the Service or deliver the deliverables, as promised. Either Party may terminate this Agreement upon such notice.
17. **Disclaimer of Warranties.** ANY DELIVERABLE WHEN ACCEPTED BY THE CLIENT IS OFFERED ‘AS IS’ AND IS DEEMED AS PROPERLY FUNCTIONING. THE PROVIDER DISCLAIMS ANY FUTURE RESPONSIBILITY DUE TO DEGRADATION, MISUSE OR ANY OTHER EVENT THAT MAY COMPROMISE THE EFFECTIVENESS OF WHAT IS DELIVERED. THE PROVIDER DISCLAIMS ANY WARRANTY AFTER THE INITIAL INTEGRATION.
18. **Limitation of Liability.** UNDER NO CIRCUMSTANCES SHALL EITHER PARTY BE LIABLE TO THE OTHER PARTY OR ANY THIRD PARTY FOR ANY DAMAGES RESULTING FROM ANY PART OF THIS AGREEMENT SUCH AS,

BUT NOT LIMITED TO, LOSS OF REVENUE, ANTICIPATED PROFIT, OR LOST BUSINESS, COSTS OF DELAY OR FAILURE OF DELIVERY, WHICH ARE NOT RELATED TO OR THE DIRECT RESULT OF A PARTY'S NEGLIGENCE OR BREACH.

19. **Severability.** In the event any provision of this Agreement is deemed invalid or unenforceable, in whole or in part, that part shall be severed from the remainder of the Agreement and all other provisions shall continue in full force and effect as valid and enforceable.
20. **Waiver.** The failure by either Party to exercise any right, power, or privilege under the terms of this Agreement will not be construed as a waiver of any subsequent or future exercise of that right, power, or privilege or the exercise of any other right, power, or privilege.
21. **Legal and Binding Agreement.** This Agreement is legal and binding between the Parties as stated above. This Agreement may be entered into and is legal and binding both throughout Europe and in the United States. The Parties each represent that they have the authority to enter into this Agreement.
22. **Governing Law and Jurisdiction.** The Parties agree that this Agreement shall be governed by the State or Country in which both Parties do business. If the Parties do business in different States and/or Countries, this Agreement shall be governed by Slovenian law.
23. **Entire Agreement and Completeness.** This Agreement is delivered as part of a "quote" or "report" document (the "Document") and may include attachments (the "Attachments") as listed in the section "attachments". This Document has 7 pages. This Agreement refers to such Document in its entirety including all Attachments and cannot be signed or interpreted without them. Both Parties hereby declare that they have read and comprehend the entire Document and all Attachments, and sign this Agreement to accept the Document in its entirety and all Attachments. The Parties acknowledge and agree that this Agreement represents the entire agreement between the Parties. If the Parties desire to change, add, or otherwise modify any terms, they shall do so in writing to be signed by both Parties.

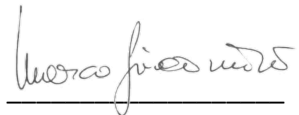
The Parties agree to the terms and conditions set forth above as demonstrated by their signatures as follows:

Mr. Claudio Nardin
CEO at Axelered Doo
(Provider)



Signed on: 2025-04-24

Marco Giacomini
CEO at Real Comm Srl
(Client)



Signed on: _____

Posture Evaluation

Technical documentation

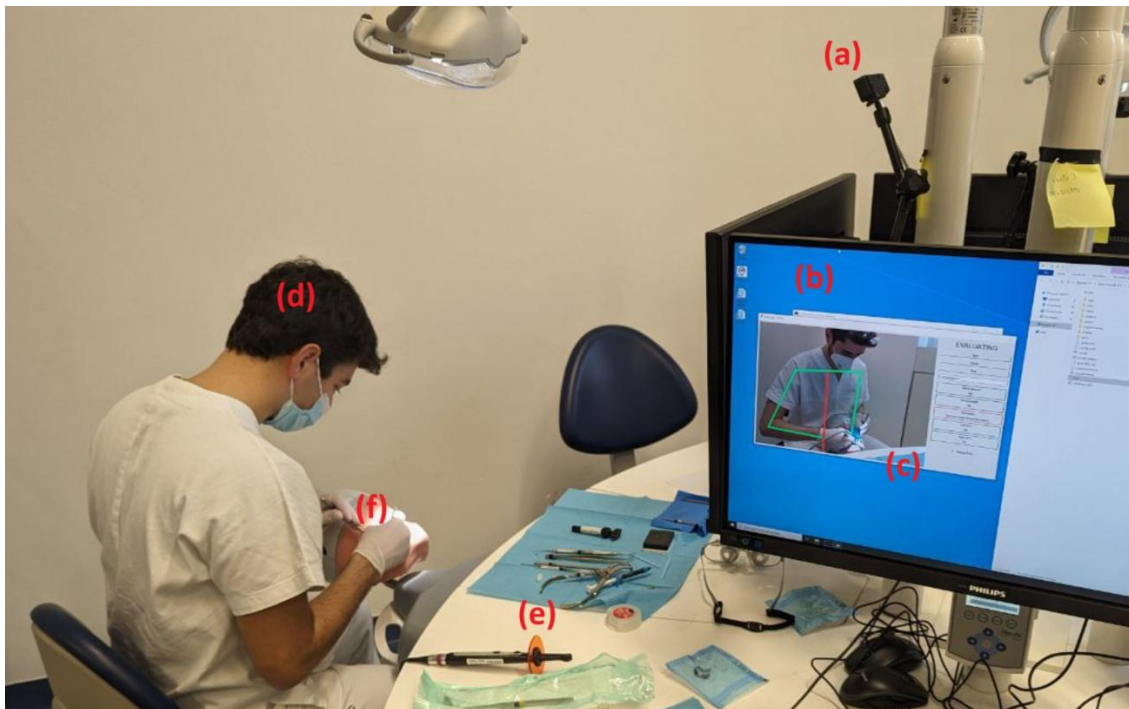
This document serves as the technical documentation for the *Posture Evaluation* project. It outlines the methodologies used to assess joint angles, movement tolerances, and overall posture analysis. The content provides a comprehensive reference for the procedures, data collection, and analysis techniques employed throughout the project, ensuring consistency and accuracy in evaluating posture-related metrics. Additionally, this document details the technical capabilities and limitations of the developed technology.

⚠ Disclaimer. *The provider does not possess specialized expertise in postural evaluation, biomechanics, or any medical implications related to posture assessment. The provider acts solely as a technology developer and supplier, with no responsibility for clinical validation. It is the sole responsibility of the client or end user to determine whether the features, specifications, and performance described herein are appropriate and sufficient for the intended use and expected outcomes.*

Revision	Referent	Date
Rev01	Ing. Svara Borut	10/10/2024
Rev02	Ing. Svara Borut	22/10/2024
Rev03	Ing. Svara Borut	18/03/2025
Approval	Ing. Svara Borut	23/03/2025
Rev04	Ing. Feresin Damian	16/04/2025
Approval	Ing. Feresin Damian	16/04/2025
Rev05	Ing. Svara Borut	18/04/2025
Approval	Ing. Svara Borut	18/04/2025

Overview

The system consists of a **sensor (a)**, an edge computing **device (b)** and a **software (c)** running on the edge device. The system is designed to be used during **dentist (d)** training. This training takes place using a **simulation device (e)**, which replicates a real workstation and includes a **mannequin (f)** that realistically represents a **patient (g)**.



The purpose of this technology is to use **sensor (a)** to capture data on the **dentist's (d)** posture and process this information via **software (c)** running on **edge device (b)**. The goal is to determine whether the training is being conducted with an ergonomically correct posture that does not contribute to potential musculoskeletal strain.

Setup & Automated Positioning

The **sensor (a)** should be positioned on the **simulation device (e)** as illustrated below, ensuring that all specified distances and angles are respected to achieve the expected measurement accuracy. The system has been tested under various conditions and has demonstrated reliable operation. However, it is not currently advised to position **sensor (a)** differently than in either **Option A** or **Option B**.

- **Option B** is optimal for exercises where the dentist (**d**) works both in front of the mannequin (**f**) and on the opposite side, offering the highest posture evaluation accuracy in both positions.
- **Option A** provides greater flexibility, allowing the dentist (**d**) to operate from any position around the mannequin. However, it is less precise when the dentist moves to the extreme sides.



Option A: Front Setup



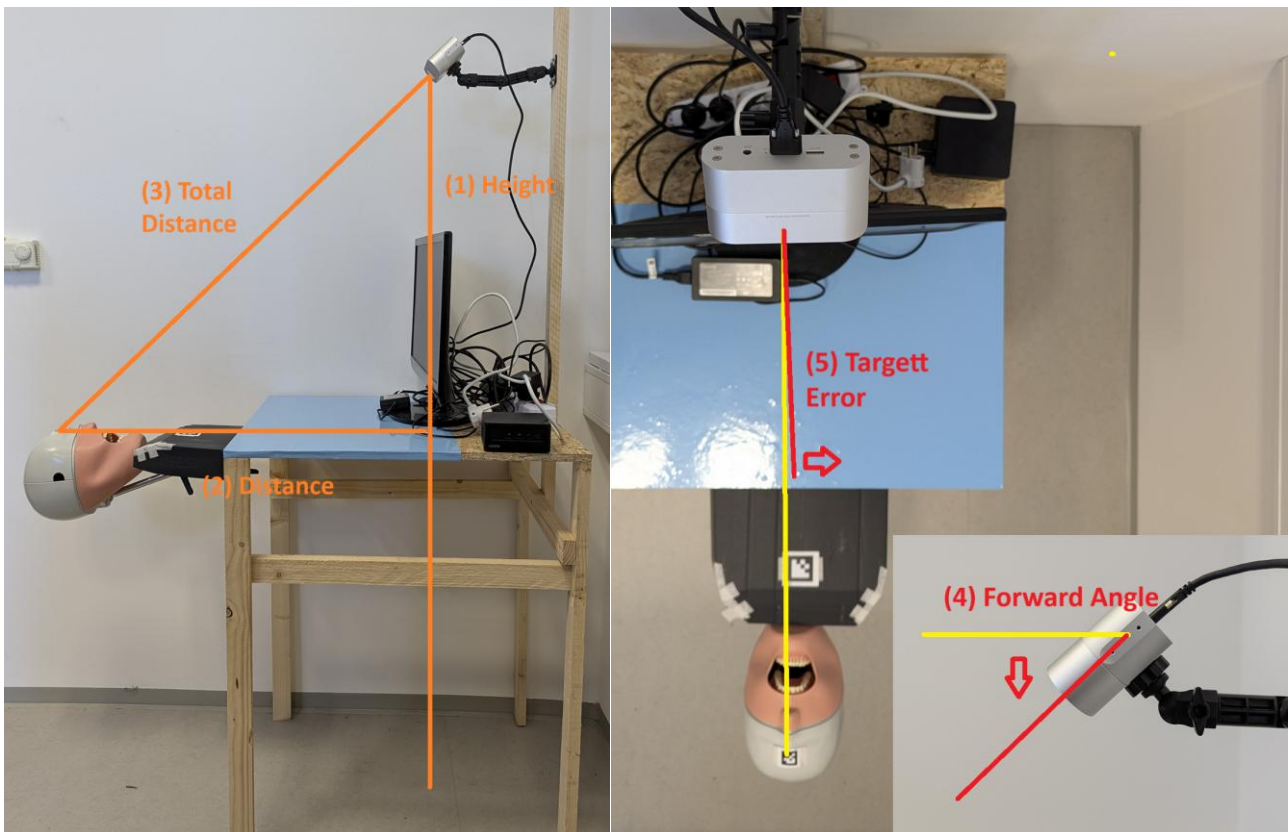
Option B: Side Setup

Refer to the following figures for details on measurement strategies.

Parameter	Measurement	Tolerance	Ref
Camera floor height	160 cm	±5 cm	(1)
Horizontal distance from the mannequin's geometric centre	65 cm	±5 cm	(2)
Total distance from the mannequin's geometric centre	100 cm	±5 cm	(3)
Forward angle	40°	±3°	(4)
Mannequin's target angle	0°	±5°	(5)
Off-front angle (Option B)	30°	±15°	(6)

The **sensor (a)** should be positioned on the **simulation device (e)** in a stable and secure manner, ensuring it is not subject to continuous external disturbances caused by the environment, the **dentist (d)**, the **patient (g)**, or any other ongoing activity near the sensor. It should be reasonably isolated from constant vibrations and sudden movements that could affect the stability of the recording.

The software performs a calibration routine upon startup and applies minor real-time corrections for small disturbances during normal operation. However, **excessive disturbances** at any stage may **compromise measurement accuracy**, rendering the evaluation ineffective.



In conjunction with software-based vibration dampening, **software (c)** also performs **global positioning correction** of **sensor (a)**. The software automatically detects the floor, its levelling axes, and its orientation. These measurements are combined to detect and correct the camera's positioning, allowing for minor misalignments from the suggested orientation.

The system also detects and informs the user if the camera has been positioned improperly and, therefore, cannot be reliably used for its intended purpose. Despite the automated adjustment routine, it is not advisable to use the system outside the recommended positioning and alignment.

Additionally, the system corrects the mannequin's positioning by detecting the broader shape of its head and mathematically adjusting the targeting. For this algorithm to function correctly, the camera must be **pointing directly at the mannequin's mouth** at the defined **sight distance (3)** when the mannequin is in its upper rest position.

Requirements

Given the specific nature of **software (c)**, the specialized **sensor (a) hardware technology** required to achieve the precision necessary for the intended application (a high-resolution ToF Depth Camera), and the AI-based algorithms adopted, the hardware requirements are stringent and must be met to ensure accurate results.

Although the technology may work on a broader range of devices, it has only been adequately tested on the **hardware specified herein**. Therefore, it should only be used with this hardware or better (in which case, additional software, libraries, and/or drivers may need to be installed to ensure backward compatibility).

Sensor (a)	Orbbec Femto Bolt 4K RGB with HDR Depth 1 Mega Pixel ToF 120deg FOV 6Dof IMU USB Type-C https://www.orbbec.com/products/tof-camera/femto-bolt/
Edge computer (b)	Asus Nuke Mini 13 16 GB RAM Intel 13 th Gen i5-1340P Intel UHD Graphics - Intel Iris Xe Graphics https://www.asus.com/displays-desktops/nucs/nuc-mini-pcs/asus-nuc-13-pro/ https://www.cpubenchmark.net/cpu.php?cpu=Intel+Core+i5-1340P&id=5237

Postural Algorithm

As defined by the client (and other consultants on its behalf), the postural evaluation algorithm is comprised of several main components that act as a processing pipeline to determine posture, output feedback to the **user (d)**, and eventually generate a **report** (as defined later in the document).

I. Posture Detection

Software (c) processes the live stream acquired by **sensor (a)** via advanced AI algorithms to detect all persons in the scene. The algorithm adopts state-of-the-art CNNs to segment the scene, detect and determine the 2D position of the user within the RGB feed, project the segmentation onto the point-cloud data from the ToF, and adopt a weighted-vote-clustering algorithm to match a normalized skeleton to the underlying 3D space. This stage yields the detection of one (or more) skeletons in the 3D space in front of the camera.

II. Correction and Filtering

When starting, the software uses a short time frame (<5 seconds) to perform a sensor-fusion algorithm that will use all **sensor (a)** internals (RGB, ToF, IMU) to detect and determine the camera positioning and alignment. The algorithm automatically detects orientation, alignment, height, and relative positioning to the **simulation device (e)** and uses such information to produce a **real-world correction matrix**. The correction matrix is used in conjunction with smoothing filters, occlusion correction kernels, and fine-tuned heuristics to correct the detections (I) and determine the **user (d)** position relative to the **simulation device (e)**. During this stage, the system also determines and tracks the primary user, ignoring any observer or other person present in the scene who is not the main actor using the **device (e)**. This stage yields a single skeleton, oriented, scaled, and corrected in respect to **device (e)**.

III. Scalar Computation

To shift the measurement into a human-understandable space and to better align with the literature available to the client's consultants, the 3D skeleton detected (II) is used to compute a series of scalars to be later used to evaluate the correctness of the posture. Such scalars are a mix of distances and angles, as described in **Appendix I**.

IV. Exercise Matching

The concept of exercise is defined as a collection of one or more distinct unordered postures. A **posture** is a **set of one or more constraints** on the computed scalars (III). **Software (c)** does not impose temporal or ordering constraints but rather constantly evaluates the current values (III) against all provided postures in the current exercise. It assumes the **closest** one (i.e., the one with the lowest error – see **V**) as the posture the user is currently intended to assume.

V. Error Determination

The identified **posture (IV)** is used to determine **user error**. Each postural constraint is comprised of multiple values:

- **Correct range (A)** – [green_lower, green_upper]
- **Warning range (B)** – [yellow_lower, green_lower] U [yellow_upper, green_upper]
- **Tolerance time (C)** – t_tolerance

The system continuously checks all constraints and expects the scalars to be within the **correct range (A)**. If the user is outside **(A)** but within the **warning range (B)** or outside the **warning range (B)**, it is considered a “warning” or an “error”, respectively. In both cases, the error is **not signalled** to the user or taken into account until the erroneous posture **persists** for more than the **tolerance time (C)** defined by t_tolerance. However, when the error is corrected or reduced in severity, feedback is given immediately (this asymmetric hysteresis improves UX).

VI. Feedback and Reporting

Detected **warnings and errors** are signalled to the user by overlaying a coloured line or dot on the erroneously positioned limb (yellow for warnings, red for errors) and by emitting sound feedback, which may be muted. All errors are collected in a time series (alongside camera frames) and are then outputted into a separate report file at the end of the exercise.

System Latency: As explicitly requested by the client to ensure real-time responsiveness the computation time is capped at **200 milliseconds** on the computational architecture (b). The computation time is defined as the time between: The latter of the RGB or Depth frame is detected by the sensor and fully transferred onto the edge computer (b) onboard RAM. And the time in which the posture skeleton is first rendered on screen excluding screen latency. The performance is intended at conventional OS Idle load (no other computationally intensive processes are active) and at 5 FPS acquisition rate of the depth channel.

Reports & Outputs

Software (c) provides multiple types of output to be shared and used by external users. These outputs are designed for specific purposes and are defined as follows:

1. **CSV Recording** – The software generates an exhaustive CSV file containing a time series (sampled once per second) of all scalars (III) and errors determined (V). This file provides a compressed yet comprehensive snapshot of the entire exercise performance for future debugging, back-testing evaluation, and research analysis. It also serves as a solid foundation for statistical assessments.
2. **JSON Report** – The JSON exercise report provides a structured and compact aggregation of the errors detected during the exercise. It is specifically designed to be used by other software, such as online platforms, front-end interfaces, and third-party interactive applications. The file contains a list of error conditions, including:
 - Start time
 - Stop time
 - Severity
 - Affected limbs list
 - Aggregated scalar values
3. **PDF Report** – The software (c) can also generate a human-readable PDF report, which includes all the information from the JSON report (2) and additionally contains frames depicting the user (d) in the erroneous position. This report is intended as an exercise outcome document to help the user self-improve and provide instructors with performance documentation.

Tuning & Customization

Regarding the postural algorithm, the processing pipeline is robust but includes multiple configurable meta-parameters. The underlying **detection algorithm (I)** has been extensively evaluated in different scenarios and is considered stable. Therefore, further tuning of this algorithm is not required.

However, the rest of the pipeline includes various filters, corrections, and adjustments (around 30 configurable parameters) that have been pre-configured based on real-world assumptions. In some cases, further refinement of these meta-parameters may be necessary to adapt to different setups.

Software Structure

⚠ Disclaimer. *The software interfaces described in this documentation, including all component methods and their specifications, are provisional design concepts only. All technical implementations remain subject to change during development without prior notice or obligation. No functionality, signature, or behavioural characteristic is to be considered final until formal release.*

The software package consists of two modular Windows Forms components written in C#, designed for embedding into third-party .NET desktop applications. Additionally, it includes a fully functional Windows desktop application that demonstrates component integration and usage, serving as both a reference implementation and testing environment.

Postural View Component

The Postural View Component captures the user's live video feed through the system camera and visually identifies postural errors in real time. This component integrates the complete machine learning (ML) pipeline, including video capture, pose estimation, and posture evaluation logic. Key features include:

- **Live Camera Feed:** Displays real-time video input from the user-facing camera
- **Pose Detection and Feedback Overlay:** Renders visual feedback directly on the live video stream, highlighting deviations from the expected reference posture
- **Self-contained ML Pipeline:** Encapsulates all required models and logic to perform real-time posture tracking and error classification without external inference engines

The component provides these control methods:

- **SetReferencePosture(Posture reference):** Loads and stores a reference posture for user evaluation
- **SetSettings(Settings settings):** Configures component behavior through a settings object (error sensitivity thresholds, visual overlay options, timing constraints)
- **StartEvaluating():** Initiates the evaluation session, comparing user posture against the reference
- **StopEvaluating():** Terminates the current evaluation session and completes posture tracking
- **GetStatus() -> Status:** Returns the component's operational state
- **GetCurrentPostureJson() -> PostureDiff** Returns the current posture object including errors in relation to the reference posture.
- **GetReportJson() -> Report:** Generates a JSON structure containing evaluated posture data (joint coordinates, error flags, metrics)
- **ProducePosturePdf() -> String:** Creates a PDF report with posture images, evaluation results, and error annotations for documentation
- **OnStatusChanged((Status) => void):** Triggers an asynchronous event whenever the component evaluating status.
- **OnCurrentPostureChanged((PostureDiff) => void):** Triggers an asynchronous event whenever the posture diff status transitions. This allows the host container to react in real time to postural changes without relying on polling mechanisms.

Exercise Editor Component

The Exercise Editor Component enables creation, visualization, and modification of postural exercises. Each exercise consists of configurable posture sequences that can be saved. The component offers both graphical and programmatic interfaces for editing the underlying JSON structure that defines complete training or evaluation routines. Key features include:

- **Exercise Definition Interface:** Builds complex postural routines through defined target postures
- **JSON-Based Configuration:** Uses structured JSON objects for exercise representation, supporting export/edit/reload functionality

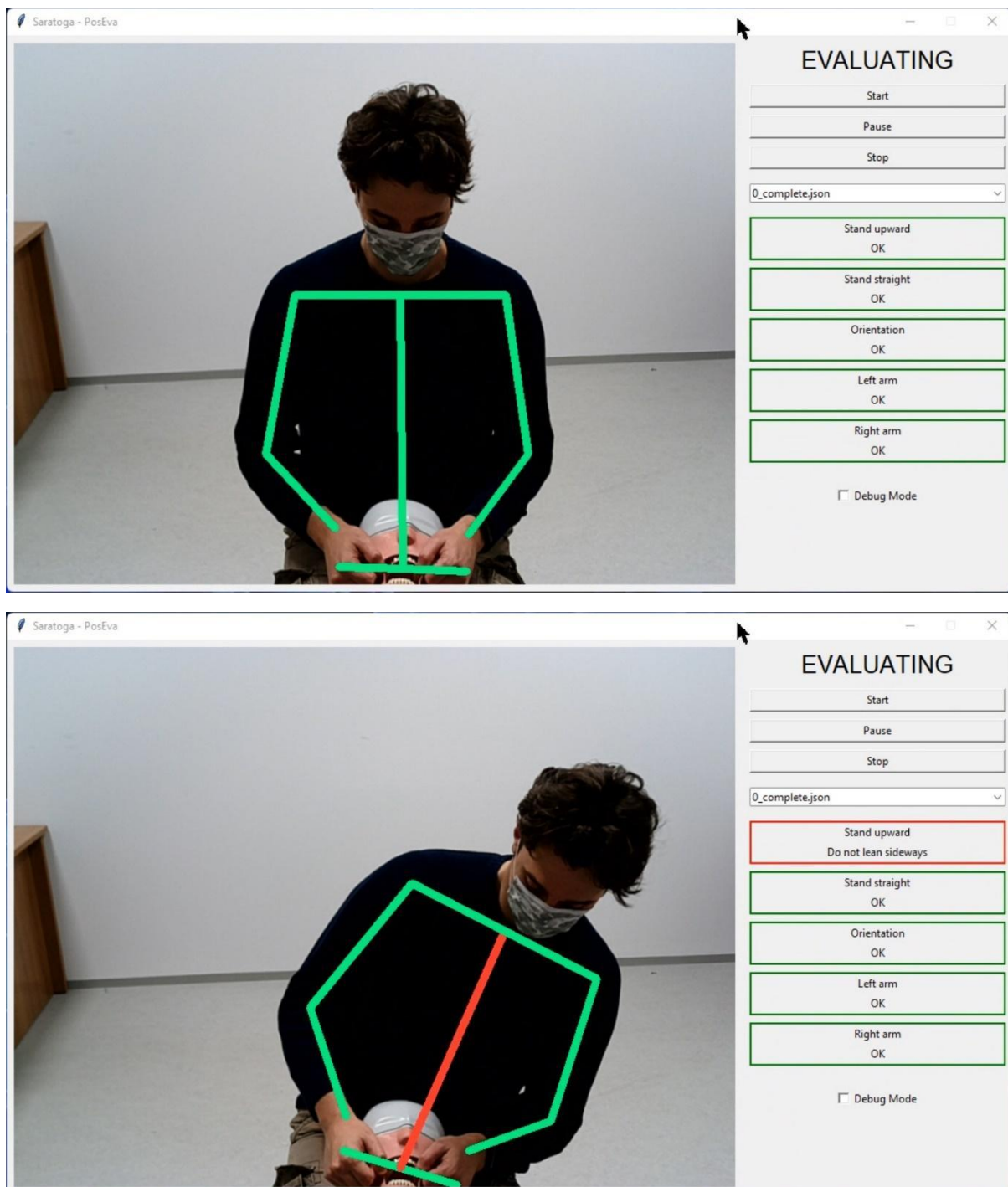
The component offers these control methods:

- **SetPosture(object posture):** Loads a posture configuration into the editor
- **GetPosture():** Returns the current posture definition as a JSON object
- **Reset():** Clears the current posture definition and restores the editor's initial state

Demonstrator Application. Alongside the reusable components, the deliverable includes a standalone Windows desktop application developed in C#. This application serves as a demonstrator and reference implementation that showcases:

- Integration of both components in a real-world application.
- A workflow for capturing a reference posture, evaluating the user's posture, and generating reports.
- Creation and editing of postural exercises using the editor component.

This application is intended to provide clients with a concrete example of how to embed and interact with the components within their own software solutions.



Appendix I: Joint Angle Tolerances

This section provides an estimation of the angular tolerances and the expected level of precision achievable with the Orbbec Femto Bolt cameras. Image 01 - Joints illustrates the range of movements for the operator, with angles labeled with codes for clarity. Each movement is systematically represented to ensure accuracy in the analysis. The specific tolerances for each movement described in the image and the accompanying table are outlined.

Definitions:

- **Image 01** depicts the scalars computed for individual joints documented in Table 01. Where only one limb is shown in the images, the movement is considered to apply to both limbs.
- **Table 01** indicates both the computed scalars and their description according to medical literature and the optimal posture as provided by the client. **Tolerance** column is the estimate of the detection precision of individual scalars as processed by the algorithm and projected onto the absolute coordinate system. **Optimal** column indicates the value of the scalar for the ideal posture, **Green**, **Yellow**, and **Grace time** are values provided for the warning and error estimation as described in "V. Error Determination" section of the postural algorithm described in this document.

Image 01 - Joints

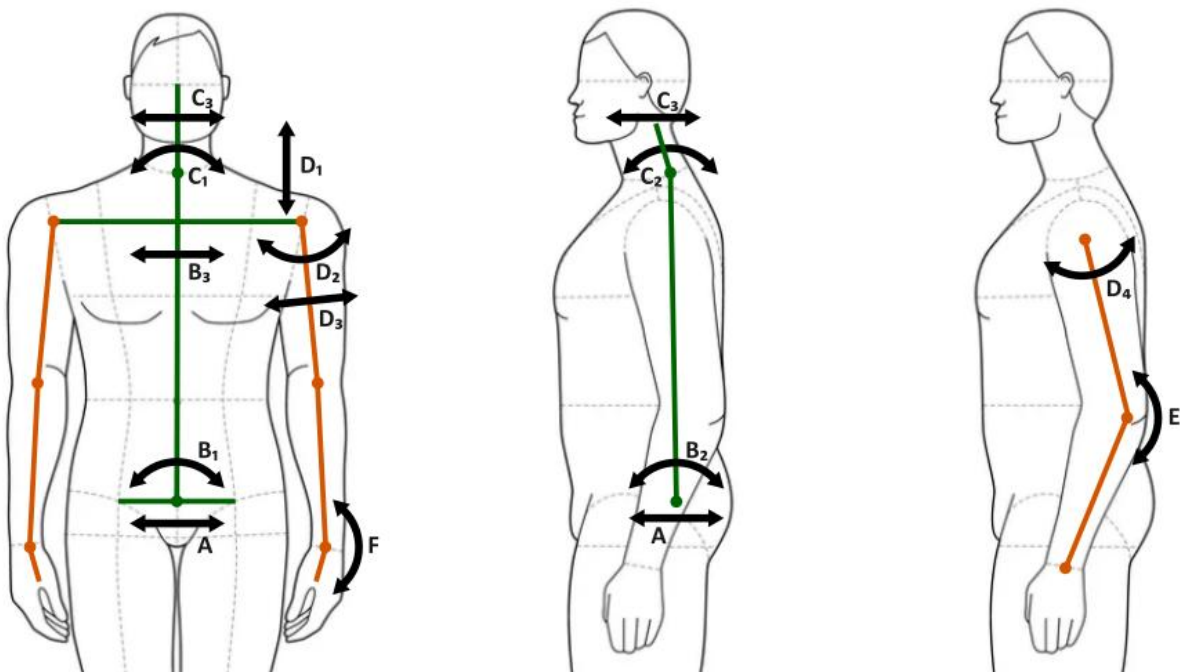


Table 01 – Joint Detection Tolerances And Optimal Dentist Position

Measure	Movement	Code	Description	Tolerance (deg)	Optimal (deg)	Green (deg)	Yellow (deg)	Grace time (sec)
Hips	User orientation	A	Is considered an absolute value that serves as a reference for the direction of the operator's body when seated in relation to the origin (the patient).	$\pm 10^\circ$	0	/	/	
Spine	Spine lateral flexion	B ₁	Refers to the side-bending motion of the vertebral column. This movement involves bending the torso to the left or right, allowing the ribs and pelvis to shift sideways.	$\pm 5^\circ$	0	5	10	
	Spine flexion	B ₂	Describes the forward and backward bending movements of the spine. Flexion involves bending forward, decreasing the angle between the vertebrae, while extension involves straightening or bending backward, increasing that angle.	$\pm 5^\circ$	0	10	25	
	Spine extension					5	10	
	Spine rotation	B ₃	Is the twisting motion of the vertebral column, allowing for the torso to turn left or right. This movement occurs in the thoracic region of the spine.	$\pm 10^\circ$	0	10	30	
Neck	Neck lateral flexion	C ₁	Involves bending the neck sideways, bringing the ear closer to the shoulder.	$\pm 5^\circ$	0	5	15	
	Neck flexion	C ₂	Describes the forward and backward bending of the neck. Flexion involves bringing the chin toward the chest, while	$\pm 5^\circ$	0	10	25	

	Neck extension		extension involves tilting the head back, moving it away from the chest.		0	5	15	
	Neck rotation	C ₃	Refers to the turning motion of the head from side to side. This movement allows for the neck to pivot, enabling individuals to look over their shoulders or scan their surroundings.	± 10°	0	10	20	
Shoulder	Shoulder elevation	D ₁	Involves the upward (elevation) and downward (depression) movement of the shoulder girdle. Elevation brings the shoulders closer to the ears, while depression lowers them back to their resting position.	± 30°	0	/	/	
	Shoulder depression					/	/	
	Shoulder abduction	D ₂	Describes the movements of lifting the arm away from the body (abduction) and bringing it back toward the body (adduction). Abduction occurs when the arm is raised to the side, while adduction happens when it is lowered back to the side.	± 10°	0	10	20	
	Shoulder adduction					20	40	
	Shoulder internal rotation	D ₃	Refers to the rotational movements of the shoulder joint. External rotation involves turning the arm outward away from the body, while internal rotation involves turning the arm inward toward the body.	± 10°	0	/	/	
	Shoulder external rotation					/	/	
	Shoulder flexion	D ₄	Involves the forward and backward movement of the arm at the shoulder joint. Flexion raises the arm forward and upward, while extension lowers it back down or moves it backward.	± 10°	0	20	50	
	Shoulder extension					10	20	
Elbow	Elbow flexion	E	Describes the bending (flexion) and straightening (extension) of the elbow joint. Flexion decreases the angle between the forearm and upper arm, while extension increases it.	± 10°	90	10	20	
	Elbow extension					10	20	

Wrist	Wrist deviation	F	Encompasses the bending movements of the wrist. Flexion involves bringing the palm closer to the forearm, while extension involves moving the back of the hand toward the forearm. Additionally, radial deviation (moving toward the thumb side) and ulnar deviation (moving toward the pinky side) allow for lateral movements of the wrist.	$\pm 30^\circ$	0	30	30	
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